BLOG-DOC-16

SEO, Search, XML Sitemap, Canonical & Redirect Specification

Wave-2 structured-data source-model reconciliation

Revision	v1.0-RC5
Date	11 September 2026
Status	OWNER-AUTHORIZED WAVE-2 RECONCILIATION
Supersedes	v1.0-RC4 only where this reconciliation explicitly says so
Authority rule	This revision resolves existing contradictions only; it creates no new product scope.

Normative reconciliation decisions

CB-W2-02 structured-data source model

The structured_data_inputs source is the canonical DOC-07 RC2 keyed object, not an array. Structured-data generation iterates deterministic object-type keys and validated field maps. Existing fallback, indexability, canonical, search, sitemap, redirect and schema.org output algorithms are otherwise unchanged.

Security boundary

Raw JSON-LD directives remain prohibited as editorial input. The generator, not the editor, owns @context/@type/output construction. Unknown or invalid source keys fail validation before generated structured data is emitted.

Invariant / count effect

No SEO/search/sitemap/redirect feature, indexability class, Page ID, ranking rule, worker job, or tenant boundary is added or removed. This revision solely eliminates the object-versus-array contradiction with DOC-07/DOC-08/DOC-15.

Incorporation rule

Supersedes only the structured-data source representation clauses in Sections 5 and 14 and related examples/appendices of RC4. All other RC4 discovery semantics remain incorporated unchanged.

Source inclusion note

The following pages are the complete inherited v1.0-RC4 source document. They remain normative except where a clause is explicitly superseded by the reconciliation pages above. This packaging avoids silently rewriting historical source pages while issuing the controlling v1.0-RC5 contract.

BLOG-DOC-16

SEO, Search, XML Sitemap, Canonical & Redirect Specification

Developer execution authority for resolved SEO output, indexability, canonical selection, structured data, breadcrumbs, public/Admin search behavior, XML sitemap discovery and redirect/SEO interaction across all 14 isolated public sites.

14	14	1	0
isolated public sites	frozen Page IDs	site-scoped search authority	cross-site public search fallbacks

CORE DISCOVERY RULE

Every SEO value, public search result, sitemap URL and redirect decision begins inside exactly one trusted site context. Route identity comes from DOC-09; public data shape comes from DOC-13; HTTP transport comes from DOC-15. DOC-16 computes discovery behavior without changing Page IDs, route families, content lifecycle, permissions or tenant ownership.

Document control	Value
Status	REVIEW CANDIDATE - prepared from finalized/frozen BLOG-DOC-01 through BLOG-DOC-15; freeze after DOC-16 approval
Version	1.0-RC4
Date	10 September 2026
Upstream authorities	BLOG-DOC-01 through BLOG-DOC-15 - finalized/frozen source-of-truth for this DOC-16 build
Primary authority	SEO fallback hierarchy; canonical-selection algorithm; robots/indexability; Open Graph; structured-data generation; breadcrumb generation; XML sitemap/index behavior; public search indexing/query/ranking/filtering/pagination; reindex/diagnostics semantics; redirect SEO interaction/conflict/loop rules.
Explicit non-authority	Media transformation/CDN (PDF-17), worker/queue/cache mechanics (PDF-18), session/credential/CSRF/CORS/security implementation (PDF-19), performance/DR targets (PDF-20), exhaustive release QA (PDF-21), repository/framework details (PDF-23), deployment mechanics (PDF-24), visual design (PDFs 25-27).

DEVELOPER LOCK TARGET

A competent developer must not have to guess which SEO field wins, whether an archive is indexable, which canonical URL to emit on page 2, how Open Graph falls back, which structured-data objects are generated, what enters a site sitemap, how a query is normalized/ranked, how ties are broken, whether reindex can cross tenants, or how manual redirects interact with canonicalization/live routes. These decisions are locked here; later documents implement infrastructure without redefining them.

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AUTHORITY BOUNDARY

THIS DOCUMENT OWNS discovery computation and SEO/search/sitemap/redirect interaction semantics. It DOES NOT create new public Page IDs, Admin pages, permissions, content fields, database columns, API endpoints or tenant models. When an upstream transport/schema gap is detected, DOC-16 flags it rather than inventing a hidden endpoint or alternate persistence path.

1. Purpose, Authority and Conflict Rule

1.1 Purpose

BLOG-DOC-16 converts the frozen routing, content, persistence, page behavior, public data and API contracts into deterministic discovery behavior. It defines how already-approved inputs become final SEO projections, crawl directives, sitemap records, search results and safe redirect outcomes for exactly one site at a time.

1.2 One source of truth per decision

Decision	Canonical authority	DOC-16 relationship
Which public pages exist / baseline indexability	DOC-05	Consumes 14 Page IDs; does not add pages.
Post/content/SEO input fields and lifecycle	DOC-07	Consumes fields/states; does not add fields/states.
Persistence / RLS / site ownership	DOC-08	Reads frozen records; no new table/column here.
Host, route, slug, 308, 404/410 and base redirect precedence	DOC-09	Consumes route canonical URL and live-route precedence; adds SEO/crawler computation only.
Public/Admin page behavior	DOC-10/11	Consumes states/actions; does not alter page flows.
Public component anatomy	DOC-12	Breadcrumb/SEO data drives components without visual redesign.
Public PageData/SeoProjection object shape	DOC-13	DOC-16 computes values into exact existing shapes.
14 frontend app boundary	DOC-14	Same algorithms are implemented/consumed across isolated frontends.
HTTP endpoints/params/status/schemas	DOC-15 RC5	DOC-16 consumes exact transport and may not add hidden API fields/endpoints.

1.3 Conflict rule

CONFLICT RULE

If a DOC-16 algorithm would require a new Page ID, new Admin module, new permission, new persisted lifecycle state, new database field, new API endpoint/schema field, cross-site ownership, or a change to DOC-09 route resolution order, implementation stops. The relevant upstream authority is reopened through controlled change; DOC-16 does not silently broaden scope.

2. Frozen Upstream Facts and Authority Chain

Authority	Frozen input consumed by DOC-16
DOC-01	14 public domains / 14 isolated public frontends / one Central Admin / shared backend and centralized PostgreSQL; shared infrastructure never means shared ownership.
DOC-02	Exact SEO permissions seo.view + seo.edit; Redirect permissions redirects.view/create/edit/delete; Search permissions search.view + search.reindex; server-side authorization and non-disclosure.
DOC-03	FC-SEO-01..15, FC-RED-01..08 and FC-SEARCH-01..08 exist; public search is same-site/public-eligible; SEO warnings are advisory.
DOC-04	Search indexes, sitemaps, canonical URLs and redirects are tenant-aware; no mixed public sitemap; missing scope fails closed.
DOC-05	14 Page IDs. Home/Article/5 Static = INDEX-ELIGIBLE; Category/Tag/Author = CONDITIONAL-INDEX; Search/Preview/404/410 = NOINDEX.

Authority	Frozen input consumed by DOC-16
DOC-06	One SEO Admin page, one Redirect module, one Search & Indexing page; sitemap automation is not a separate Admin page.
DOC-07	Post SEO inputs: slug, seo_title, meta_description, canonical_override, index_policy, link_robots_policy, og_title, og_description, og_image_media_id, structured_data_inputs, breadcrumb_label.
DOC-08	Categories persist seo_title/meta_description/canonical_override/index_policy; Authors persist same; Tags do not persist custom SEO fields; site_settings.seo_defaults is JSON whose semantics are DOC-16 authority; redirects persist site-owned source/destination/status.
DOC-09	HTTPS/canonical host, route families, slug normalization, system 308, live-route-wins, manual redirect model, max 3 internal redirect edges, URL history is evidence not redirect, 404/410 rules.
DOC-10	SEO Admin section order and capabilities; Search diagnostics/reindex behavior; Redirect editor must validate current-route conflict, cycle and destination eligibility.
DOC-11	Search/Preview/system SEO state behavior; pagination/query scope safety; redirect body not rendered; conditional archive indexability left to DOC-16.
DOC-12	Breadcrumb/SEO components consume semantic fields; visual design remains site-local.
DOC-13	Exact SeoProjection/OpenGraph/Robots/Breadcrumb/SearchPageData shapes; optional absence is omission; arrays ordered; search result ranking order is authoritative.
DOC-14	All 14 frontends consume the same semantic contracts without site-to-site application imports.
DOC-15 RC5	/api/v1/public/page transport, SEO/redirect/search Admin endpoints, page/page_size/q limits, 202 reindex, exact request/response schemas and rate/idempotency/concurrency rules.

3. Discovery Architecture Doctrine and Identifiers

Principle ID	Locked rule
SEO16-PR-01	Discovery is computed after trusted SiteContext and route/public-eligibility resolution.
SEO16-PR-02	Route canonical identity and HTML canonical selection are separate; DOC-09 owns the former and DOC-16 the latter.
SEO16-PR-03	NOINDEX is a crawl/indexing decision, not authorization and not a substitute for hiding private content.
SEO16-PR-04	Public site search is not a network search. Every query is tenant-scoped before matching/ranking.
SEO16-PR-05	Search index state is rebuildable derived state; PostgreSQL/content authorities remain canonical truth.
SEO16-PR-06	Sitemap inclusion is stricter than page existence: only final indexable canonical public URLs are emitted.
SEO16-PR-07	Canonical override never changes routing, ownership, publication state or redirect behavior.
SEO16-PR-08	Manual redirects do not grant content access and never outrank an existing current canonical public route.
SEO16-PR-09	Generated structured data is allowlisted output derived from trusted projections; raw Admin JSON-LD is never emitted.
SEO16-PR-10	All ordering has deterministic tie-breakers; database row order is never discovery order.

3.1 DOC-16 identifier families

Family	Meaning	Example
SEO16-*	SEO/canonical/robots/OG/structured-data rule	SEO16-CAN-04
SM16-*	Sitemap/crawler-discovery rule	SM16-ELIG-03
SRCH16-*	Search indexing/query/ranking rule	SRCH16-RANK-07
RED16-*	Redirect integration/conflict rule	RED16-VAL-06
TR16-*	Traceability requirement	TR16-021

Family	Meaning	Example
QA16-*	Release acceptance assertion	QA16-SRCH-018

4. Trusted SiteContext and 14-Site Isolation

PERMANENT TENANT RULE

The same URL path, query string, slug or search phrase may exist on multiple sites. Every SEO resolver, search query, sitemap build and redirect lookup first binds to exactly one trusted site identity. No later ranking or canonical rule may broaden that scope.

Operation	Required context	Forbidden behavior
Resolve SEO for public PageData	CTX-PUBLIC-SITE or CTX-PREVIEW-SITE	Client-provided site_id, mixed-site data, cross-site fallback.
Build sitemap/robots.txt	One ACTIVE canonical site	Network-wide public sitemap; alias-as-second-site output.
Public search	One ACTIVE CTX-PUBLIC-SITE	Search all sites then filter in browser.
Admin search/reindex	One CTX-ADMIN-SITE + permission	Reindex without site identity or read another site draft.
Redirect lookup	site_id + normalized source path	Global source-path table lookup.
Structured data	One same-site PageData graph	Borrow author/media/navigation/site data from another tenant.

5. SEO Input/Source Model and Normalization

5.1 Source precedence classes

Class	Examples	Authority
Explicit object inputs	post/category/author seo_title, meta_description, canonical_override, index_policy; post OG fields	DOC-07/08
Site SEO defaults	site-level title/meta template, index/follow default, default social image, structured-data source inputs	DOC-08 site_settings.seo_defaults + DOC-15 SiteSeoInput; semantics fixed here
Canonical content fields	title/name/display_name/excerpt/body/bio/description/static content	DOC-07/08/13
Route identity	canonical host/path, page type, slug, pagination state	DOC-09/15
Derived public output	SeoProjection/OpenGraph/Robots/BreadcrumbTrail/ structured_data	DOC-13 shape; values fixed here

5.2 String normalization for SEO computation

1. Trim Unicode leading/trailing whitespace.
2. Collapse internal whitespace runs to one space for generated title/description/breadcrumb labels; do not rewrite authored punctuation or script.
3. Treat empty-after-trim values as absent for fallback purposes.
4. Never silently truncate canonical URLs, slugs or authored visible labels.
5. Generated meta descriptions may be clipped only by the explicit plain-text fallback rule in Section 8; explicit authored meta_description is preserved as authored after whitespace normalization.

6. No SEO computation transliterates, translates, stems or changes the content language.

5.3 Site-level SEO template semantics

DOC-15 SiteSeoInput.seo_title and meta_description are interpreted here as the editable site-level fallback templates used by FC-SEO-13/14. This adds semantics to existing fields; it does not add transport fields.

Field	Allowed tokens	Validation / fallback
Site seo_title	{page_title}, {site_name}	If configured, must contain {page_title} exactly once. Unknown tokens -> 422 on edit. If absent, built-in template is {page_title}.
Site meta_description	{page_summary}, {site_name}	Unknown tokens -> 422. If rendered result has no nonblank page_summary, the template is skipped and meta_description may be omitted.
index_policy	INHERIT, INDEX, NOINDEX	Applies as site-level index request; hard Page-ID NOINDEX and eligibility gates cannot be overridden.
link_robots_policy	INHERIT, FOLLOW, NOFOLLOW	Default link-follow request for ordinary public pages; Preview always nofollow.
default_social_image_media_id	same-site UUID	Must resolve to public-safe same-site media before it may become OpenGraph.image.
structured_data_inputs	validated typed source array	May supplement allowlisted generated schemas only; no @context/@type/raw JSON-LD directives from Admin input.

6. Resolved SEO Fallback Hierarchy

Page type	Resolved title source	Meta source	Canonical source	Index policy source
Home	site title template with page_title=site.display_name	site meta template; page_summary from approved site description if available, otherwise omit	DOC-09 Home route canonical	DOC-05 baseline + site index policy
Article	post.seo_title -> post.title, then site title template	post.meta_description -> post.excerpt -> generated body summary -> site meta template	valid post canonical_override -> route canonical	DOC-05 baseline + site + post index policy
Category	category.seo_title -> category.name, then site title template	category.meta_description -> category.description -> site meta template	valid category canonical_override -> route canonical	conditional gate + site + category index policy
Tag	tag.name, then site title template	tag.description -> site meta template	route canonical only	conditional gate + site policy; no tag-local index field exists upstream
Author	author.seo_title -> display_name, then site title template	author.meta_description -> bio summary -> site meta template	valid author canonical_override -> route canonical	conditional gate + site + author index policy
Search	Search title based on static Page Intro + site template	omit unless fixed approved Search intro summary exists	clean /search route canonical; q/page never canonical identity	hard NOINDEX
Five Static	StaticContent.title + site template	intro_text -> first section summary -> site meta template	DOC-09 fixed static route canonical	DOC-05 baseline + site policy
Preview	Preview display title only; private semantics	optional preview summary; never used for indexing	OMIT canonical_url	hard NOINDEX + nofollow
404/410	System message title + site template when site known	normally omit	OMIT canonical_url	hard NOINDEX

7. Document Title Algorithm

SEO16-TITLE-01

```
page_title = explicit object seo_title when supported and nonblank
             else canonical visible page title/name/display_name
template = normalized site seo_title template or "{page_title}"
if page_title is blank: use site.display_name for Home/system only; otherwise data error
resolved = render(template, page_title, site.display_name)
collapse whitespace; preserve language/script; do not auto-keyword-insert
Search/Preview/404/410 remain NOINDEX regardless of title
```

- The site template is applied after object-level explicit title fallback so editors can control the semantic page title while preserving a consistent site pattern.
- An explicit Post/Category/Author seo_title is literal content, not a nested template.
- If a configured site template contains an unknown token or omits {page_title}, the configuration is invalid and must not be accepted by the SEO edit endpoint.
- SEO warnings may flag unusually long output, but the runtime must not silently alter an explicit valid title to chase ranking heuristics.

8. Meta-Description Algorithm

8.1 Plain-text summary extraction

7. Use the explicit meta_description when the page type owns one and it is nonblank.
8. Else use the page-specific summary source: Post excerpt, Category description, Tag description, Author bio, Static intro_text.
9. For Article only, if both explicit meta_description and excerpt are absent, derive text from ArticleBody in block order using textual content only; ignore media URLs, embed code, tables used solely as layout, and source-link URLs unless their visible label is text.
10. Normalize whitespace and take the first complete sentence ending at or before 180 Unicode characters when possible; if no sentence boundary exists, take the first 180 characters and trim trailing whitespace. This generated clipping rule never changes stored content.
11. Feed that page_summary into the optional site meta_description template. If the template requires {page_summary} and no summary exists, omit meta_description rather than fabricate marketing copy.

NO FABRICATED COPY

DOC-13 explicitly forbids fabricated marketing copy when an optional SEO value is missing. The fallback engine may summarize existing approved content; it may not invent claims, keywords, offers or astrology assertions not present in the page data.

9. Canonical URL Selection and Override Validation

9.1 Default selection

SEO16-CAN-01

```
default_url = DOC09.route_canonical_url(page, canonical_host, stored_slug_or_static_key)
if page is Preview or 404 or 410: canonical_url = OMIT
elif page is Search: canonical_url = https://canonical_host/search
elif pagination.page > 1 on Category/Tag/Author: canonical_url = base archive canonical (Section 12)
elif supported explicit canonical_override is present and passes SEO16-CAN-VALIDATE:
    canonical_url = normalized override
else: canonical_url = default_url
```

9.2 Canonical override validation

Rule	Required behavior
CAN-V01	Absolute HTTPS URL only.

Rule	Required behavior
CAN-V02	No username/password authority component.
CAN-V03	No fragment; a submitted fragment makes the value invalid rather than silently becoming routing identity.
CAN-V04	Host must be syntactically valid after IDN/hostname normalization; later security policy may block additional hosts.
CAN-V05	If override points to this platform site through an active alias, normalize its host to the current canonical domain.
CAN-V06	If same-site path is Preview, Search query, 404/410 system identity, or a currently nonpublic site-owned entity, reject.
CAN-V07	External HTTPS canonical is permitted because FC-SEO-04 exists, but it never grants cross-site access, changes routing or creates syndication.
CAN-V08	Canonical override is not a redirect and never creates/edits a redirect row.
CAN-V09	When an override is external, the local URL is excluded from XML sitemap eligibility to avoid declaring a different canonical destination.
CAN-V10	Duplicate/restore/publish semantics remain DOC-07; DOC-16 only validates/uses the current approved input.

10. Robots/Indexability and Follow Algorithm

SEO16-ROBOTS-01

```
if PageID in {SEARCH, PREVIEW, 404, 410}: index = false
else if site.status != ACTIVE or public response != 200: index = false
else if site.index_policy == NOINDEX: index = false
else if page is Article and post.index_policy == NOINDEX: index = false
else if page is Category and category.index_policy == NOINDEX: index = false
else if page is Author and author.index_policy == NOINDEX: index = false
else if PageID is CONDITIONAL-INDEX: index = conditional_gate(page)
else: index = true

follow = false for Preview
follow = false if site.link_robots_policy == NOFOLLOW
follow = false for Article if post.link_robots_policy == NOFOLLOW
otherwise follow = true (including Search/404/410 when site context exists)
```

- Explicit INDEX may request indexing only after hard eligibility gates pass; it cannot override Search/Preview/404/410 NOINDEX or a disabled/nonpublic resource.
- NOINDEX does not remove a page from site-internal search by itself; public search visibility is based on public eligibility, not crawler policy.
- NOFOLLOW is link-crawl guidance only and never changes navigation, authorization or visibility to a human visitor.

11. Conditional-Index Rules for Category, Tag and Author

DOC-05 deliberately leaves Category/Tag/Author as CONDITIONAL-INDEX. DOC-16 freezes the per-instance gate without adding new persisted fields.

Gate	Category	Tag	Author
Entity eligible	ACTIVE/public-eligible	ACTIVE/public-eligible	ACTIVE/public-eligible
HTTP/PageData state	200 READY or valid EMPTY	200 READY or valid EMPTY	200 READY or valid EMPTY
Public eligible Article count	>= 1 required for index=true	>= 1 required for index=true	>= 1 required for index=true
Object-local NOINDEX	category.index_policy may force false	No tag-local index field exists	author.index_policy may force false
Site NOINDEX	forces false	forces false	forces false
Pagination >1	noindex,follow	noindex,follow	noindex,follow
Final gate	all hard gates pass	all hard gates pass	all hard gates pass

EMPTY ARCHIVE RULE

An existing ACTIVE archive with zero articles remains a valid public EMPTY page under DOC-11/13, but its final robots.index is false and it is excluded from XML sitemap until at least one public-eligible Article is present.

12. Pagination SEO Rules

Surface	page=1	page>1	Canonical	Indexability
Category/Tag/Author	Normal archive candidate	Valid paginated archive when results exist	page=1 -> clean base URL; page>1 -> clean base URL	page=1 uses conditional gate; page>1 = noindex,follow
Search	NOINDEX results/NO_QUERY	NOINDEX results	always clean /search without q/page	always noindex,follow
Invalid/out-of-range page	DOC-15 400 for invalid syntax; valid numeric page with no result follows page flow contract	No invented redirect to last page	No indexable body	noindex/system outcome
Sitemap	Only base archive URL can be emitted	Never emit pagination URLs	N/A	base must pass final index gate

The choice to noindex archive pages beyond page 1 avoids creating query-string discovery identities while preserving human navigation and followable links. Article URLs themselves remain independently indexable and sitemap-eligible.

13. Open Graph Resolution

Field	Article	Other ordinary public pages
title	post.og_title -> resolved SEO title	resolved SEO title
description	post.og_description -> resolved meta_description	resolved meta_description when present
url	resolved canonical_url when present	resolved canonical_url when present
type	ARTICLE	WEBSITE
image	same-site post.og_image -> featured_media -> site default social image	site default social image
absence behavior	If no valid same-site media exists, omit image	If no valid site default exists, omit image

- OpenGraph.image uses DOC-13 PublicMedia and therefore exposes no storage key/private URL.
- A broken/missing image never causes cross-site fallback.
- External canonical override may be used as OpenGraph.url, but the image still follows same-site media safety.

14. Structured-Data Generation and Safe Source Inputs

14.1 Generated object allowlist

Page class	Generated JSON-LD object(s)	Required source
Home	WebSite	SiteIdentity, canonical Home URL; optional same-site logo.
Article	Article; BreadcrumbList when breadcrumb trail present	ArticlePageData, author when present, dates, canonical URL, image when present.
Category/Tag	CollectionPage; BreadcrumbList when present	Archive title/description/canonical and breadcrumb trail.
Author	ProfilePage + Person; BreadcrumbList when present	AuthorDetail + canonical URL.

Page class	Generated JSON-LD object(s)	Required source
Static	WebPage; BreadcrumbList when present	StaticContent + canonical URL.
Search	[]	NOINDEX; v1 emits no SearchResultsPage schema.
Preview	[]	Private/noindex.
404/410	[]	System/noindex.

14.2 Article JSON-LD mapping

Property	Source / rule
@context	Generated constant https://schema.org; never Admin-authored.
@type	Article
headline	Resolved SEO title or Article title; not a raw site template.
description	Resolved meta_description when present.
url / mainEntityOfPage	Resolved canonical_url when present.
datePublished	first_published_at / published_at from approved public Article projection when present.
dateModified	updated_at when present.
author	Person with display_name and same-site public author URL when available.
image	PublicMedia.url from selected OG/featured media when present.
inLanguage	SiteIdentity.default_language/locale_code when available.

14.3 Safe supplemental source-input policy

- Admin structured_data_inputs may supply only allowlisted supplemental scalar/array values for a recognized generated object type.
- Keys beginning with @ are never accepted from Admin input; @context and @type are generated by the server.
- Supplemental values cannot override canonical URL, site identity, author ownership, publication state, route identity or generated breadcrumb position.
- Unknown structured-data input type is rejected at edit validation rather than passed through to public JSON-LD.
- Raw JSON strings containing script tags, JSON-LD documents or nested arbitrary objects are prohibited by DOC-07/15 and remain prohibited here.

15. Breadcrumb Generation and BreadcrumbList Output

15.1 Visible BreadcrumbTrail algorithm

Page	Trail
Home	Omit breadcrumbs.
Article	Home -> primary Category when one exists -> Article current. If no category, Home -> Article current.
Category	Home -> Category current. Parent-category hierarchy is not expanded in v1 unless the frozen public IA exposes it; DOC-16 does not invent multi-level taxonomy UI.
Tag	Home -> Tag current.
Author	Home -> Author current.
Search	Home -> Search current when DOC-12 renders breadcrumbs.
Static	Home -> Static page current.

Page	Trail
Preview	Home destination may be omitted from private trail; current Preview label only when DOC-12 requires.
404/410	Omit or Home -> system current only when site is known and DOC-12 renders the trail.

15.2 Label fallback

12. Article current label = post.breadcrumb_label if nonblank, else Article title.
13. Category current label = category.name.
14. Tag current label = tag.name.
15. Author current label = author.display_name.
16. Static/Search/System labels come from the frozen Page Intro/static content contract.
17. Exactly one final item has is_current=true and omits destination, preserving DOC-13.

15.3 BreadcrumbList JSON-LD

When a BreadcrumbTrail is present on an indexable ordinary public page, generate BreadcrumbList with ListItem.position starting at 1 in exact visible trail order. Non-current items use resolved same-site destination URLs; current item uses the page canonical URL when present. No cross-site breadcrumb node is permitted.

16. Image SEO Boundary and Media Fallbacks

DOC-16 owns which approved media becomes SEO/Open Graph output; PDF-17 later owns image processing, variants, CDN and delivery. Alt/caption/credit authorship remains Media/Post authority.

Concern	DOC-16 rule
Featured/OG selection	Use object-specific OG media first, then Article featured media, then site default social image.
Same-site requirement	All selected media IDs re-resolve inside owning SiteContext before use.
Missing media	Omit image field; never borrow another site asset by filename/title.
Alt text	Public renderer uses authored PublicMedia.alt_text when present; DOC-16 does not fabricate alt text.
Sitemap image extension	Not required in v1; XML URL sitemap remains URL-focused unless formally added later.
Structured data image	Use public-safe resolved URL only; storage/object keys never appear.

17. Page-Type SEO Outcome Matrix

Page ID/class	Baseline	Final index rule	Canonical	Structured data	Sitemap
PUB-HOME-01	INDEX-ELIGIBLE	Index unless site hard noindex	Home canonical	WebSite	Sitemap yes if index=true
PUB-ARTICLE-01	INDEX-ELIGIBLE	Index unless site/post noindex or nonpublic	Override if valid else route	Article + BreadcrumbList	Yes if index=true and canonical is local route
PUB-CATEGORY-01	CONDITIONAL-INDEX	Need >=1 eligible article + policy gates	Override if valid else route	CollectionPage + BreadcrumbList	Base URL only if index=true/local canonical
PUB-TAG-01	CONDITIONAL-INDEX	Need >=1 eligible article + site gate	Route only	CollectionPage + BreadcrumbList	Base URL only if index=true
PUB-AUTHOR-01	CONDITIONAL-INDEX	Need >=1 eligible article + policy gates	Override if valid else route	ProfilePage + Person + BreadcrumbList	Base URL only if index=true/local canonical
PUB-SEARCH-01	NOINDEX	Always noindex,follow	Clean /search	[]	Never
Five PUB-STATIC-*	INDEX-ELIGIBLE	Index unless site noindex	Fixed DOC-09 route	WebPage + BreadcrumbList	Yes if index=true
PUB-SYS-PREVIEW-01	NOINDEX	noindex,nofollow	Omit	[]	Never
PUB-SYS-404-01	NOINDEX	noindex,follow when site known	Omit	[]	Never

Page ID/class	Baseline	Final index rule	Canonical	Structured data	Sitemap
PUB-SYS-410-01	NOINDEX	noindex,follow when site known	Omit	[]	Never

18. robots.txt Crawler-Control Behavior

NON-PAGE RESOURCE

/robots.txt is a site-scoped crawler resource, not a DOC-05 Page ID and not a CMS page. It is generated only after DOC-09 host canonicalization and trusted site resolution.

SEO16-ROBOTS-TXT-01

```
if host resolves to ACTIVE site and site.index_policy != NOINDEX:
    User-agent: *
    Disallow: /_preview/
    Sitemap: https://{canonical_domain}/sitemap.xml
elif host resolves to ACTIVE site and site.index_policy == NOINDEX:
    User-agent: *
    Disallow: /
    Sitemap: https://{canonical_domain}/sitemap.xml
else:
    no branded/site-specific robots output; routing availability follows DOC-09
```

- Search remains meta noindex rather than being hidden behind a robots.txt disallow rule, so crawlers that reach it can observe noindex.
- Preview is both meta noindex/nofollow and disallowed by robots.txt; authorization still protects Preview because robots.txt is not security.
- The Sitemap directive always uses the current canonical domain, never an alias host.

19. XML Sitemap Architecture and Resource Namespace

DOC-16 defines crawler resources without creating public Page IDs. These resources are generated per site and are not editable content pages.

Resource	Purpose	Contents
/sitemap.xml	Sitemap index	References non-empty site-local child sitemaps only.
/sitemaps/pages.xml	Fixed/indexable non-entity pages	Home + eligible fixed static URLs.
/sitemaps/articles-{n}.xml	Article shards	Final sitemap-eligible Article canonical route URLs.
/sitemaps/categories-{n}.xml	Category archive shards	Final indexable base Category URLs.
/sitemaps/tags-{n}.xml	Tag archive shards	Final indexable base Tag URLs.
/sitemaps/authors-{n}.xml	Author archive shards	Final indexable base Author URLs.

RESOURCE PRECEDENCE

Crawler-resource paths are system-owned SEO resources, not manual redirect sources or content Page IDs. Implementations must reserve /robots.txt, /sitemap.xml and /sitemaps/ from editorial redirect creation. This is a DOC-16 discovery-resource rule and does not alter DOC-09 Page ID route families.

20. XML Sitemap Eligibility, lastmod and Sharding

20.1 URL eligibility algorithm

SM16-ELIG-01

```
include(url) only if:
  site is ACTIVE and url belongs to current canonical domain
  resolved public page returns 200 and is public-eligible
  final robots.index == true
  canonical_url is present
  canonical_url == DOC09 route canonical URL on this canonical domain
  page is not Search/Preview/404/410
  pagination page == 1
else exclude
```

20.2 lastmod rules

URL class	lastmod source
Home	max of latest public-affecting Home/navigation/site-settings update known to the projection; omit when no trustworthy timestamp is available.
Article	current published Article updated_at; use publication timestamp if updated_at unavailable.
Category/Tag/Author	max(entity.updated_at, newest eligible Article.updated_at in that archive); omit if source cannot be computed reliably.
Five Static	Omit unless the approved static source has an authoritative modification instant.

20.3 Sharding and ordering

- Maximum 25,000 URL entries per child sitemap shard.
- Within each child sitemap, order by normalized canonical URL ascending.
- Shard number begins at 1 and is assigned after sorting; a content update that changes only lastmod does not change shard membership unless URL ordering changes.
- The sitemap index lists child sitemap URLs by fixed type order: pages, articles, categories, tags, authors; within type, shard number ascending.
- Do not emit changefreq or priority in v1.

21. Public Search Index Document Model

The public search index is derived/rebuildable state. It contains only same-site publicly eligible Articles and enough normalized text/identity to rank results; final ArticleSummary objects are re-projected from current canonical data rather than returned as raw index documents.

Logical field	Purpose	Public index rule
site_id	Tenant partition key	Required; server-derived; never returned to browser as authority.
post_id	Stable source identity	Required; same-site Article lookup key.
title_text	Primary search text	From Post title.
seo_title_text	Secondary search text	From explicit Post seo_title when present.
excerpt_text	Summary search text	From Post excerpt.
body_text	Rendered textual body search text	Extract visible text from approved ArticleBody blocks; no raw HTML/scripts.
category_terms	Taxonomy terms	Current same-site Category names attached to Post.
tag_terms	Taxonomy terms	Current same-site Tag names attached to Post.
author_text	Attribution term	Current same-site Author display name when present.

Logical field	Purpose	Public index rule
slug_text	Locator term	Current normalized slug.
first_published_at	Tie-break/freshness metadata	Current Post publication instant.
updated_at	Deterministic update metadata	Current durable Post update instant.

21.1 Public inclusion rule

- Only PUBLISHED and otherwise public-eligible Posts of the resolved site are in the public index.
- Post robots NOINDEX does not remove a Post from internal site search; robots controls external crawler indexing, not visitor discovery.
- Archive/Trash/Permanent Delete removes the public search document. Scheduled/Draft never enters it before actual publish.
- Search index storage may use any later-approved engine; DOC-16 freezes logical content and behavior, not vendor/technology.

22. Query Normalization and Tokenization

SRCH16-NORM-01

```
input q: DOC15 max 500 characters
1. Unicode normalize NFKC
2. trim outer whitespace; collapse internal whitespace
3. Unicode case-fold for matching
4. convert punctuation/separators to spaces except combining marks inside letters
5. tokenize into contiguous Unicode letter/digit sequences
6. preserve original script; no transliteration, stemming, synonym expansion or stop-word removal in v1
7. deduplicate tokens for coverage scoring but keep normalized full query for phrase scoring
8. if no tokens remain -> SearchPageData.NO_QUERY + [] + no pagination
```

- Normalization is identical for indexing and querying.
- A query does not interpret boolean operators, regex, wildcard, field syntax or raw search-engine query DSL.
- Changing q resets page to 1 under DOC-11/13/15.

23. Public Search Ranking Algorithm

Ranking is deterministic and language-neutral. It does not use cross-site popularity, hidden personal profiles or paid placement.

Signal	Score contribution
Normalized full query equals title_text	+160
Normalized full query appears as contiguous phrase in title_text	+110
All unique query tokens appear in title_text	+70
title token coverage ratio	+40 x coverage
Full query phrase in seo_title_text	+45
seo_title token coverage	+20 x coverage
Full query phrase in excerpt_text	+30
excerpt token coverage	+18 x coverage
category/tag token coverage	+18 x max(category_coverage, tag_coverage)
author token coverage	+14 x coverage
slug token coverage	+10 x coverage
body token coverage	+8 x coverage

SRCH16-RANK-01

```
coverage(field) = matched_unique_query_tokens / total_unique_query_tokens
score = sum(all applicable contributions)
candidate must match at least one indexed field token or exact phrase
order by score DESC, then first_published_at DESC NULLS LAST, then post_id ASC
```

RANKING IS SITE-LOCAL

No score, popularity signal, query log or result from another site may participate in ranking. Same titles/slugs on SITE-01 and SITE-02 are independent search documents and independent query sets.

24. Public Search Filtering, Pagination and Deterministic Ordering

Concern	Locked behavior
Tenant filter	site_id = trusted CTX-PUBLIC-SITE before text matching.
Eligibility	PUBLISHED + current public eligibility before ranking.
q	Normalized under Section 22; blank -> NO_QUERY, never all Posts.
page	1-based; DOC-15 default 1; invalid <1 -> 400.
page_size	DOC-15 default 20, maximum 50.
Offset	offset=(page-1)*page_size over the deterministic ranked result set for that request.
Result object	Return DOC-13 ArticleSummary in ranking order; do not expose internal score/debug payload publicly.
Pagination	Emit DOC-13 Pagination only when there is a valid previous/next/visible page set; links preserve normalized q and current site.
SEO	Search always robots.index=false; canonical clean /search; structured_data=[].

25. Admin/Editorial Search Behavior

25.1 Admin Post Search

API15-S-090 remains the transport authority: q, optional status, page/page_size. DOC-16 freezes matching/ranking inside the already-authorized site result set.

Priority	Match
1	Exact normalized title
2	Exact normalized slug
3	Title phrase
4	All query tokens in title
5	Title token coverage
6	Excerpt/body/taxonomy/author token coverage
Tie	updated_at DESC, then post_id ASC

- Status filter, when supplied, is applied before ranking and may select any of the five DOC-07 states that the authorized Admin endpoint exposes.
- Admin search may include nonpublic editorial states only after posts.view + trusted CTX-ADMIN-SITE authorization. Search.view alone does not grant Post content visibility.

- Internal-link and related-post candidate endpoints remain same-site; DOC-16 ranks only candidates the upstream content/permission rules allow and does not broaden target eligibility.

25.2 Search filters scope

DOC-15 RC5 exposes q + status as the explicit Admin Post Search filter transport. DOC-16 therefore does not invent hidden category/author/date query parameters. Additional filter transport would require a controlled DOC-15 revision.

> **ADMIN SEARCH IMPLEMENTATION BOUNDARY**
>
> DOC-16 fixes Admin Post Search matching/order semantics inside the authorized same-site candidate set, but it does not mandate a separate derived Admin search index. DOC-23 may implement that search over authoritative persistence or an approved same-site acceleration structure. The mandatory derived index defined by Sections 21-24 is the public search index.

26. Indexing Lifecycle, Incremental Updates and Reindex Semantics

Event	Public search effect	Sitemap/SEO effect
Create Draft / autosave	No public index document	No sitemap entry
Schedule	No public index document until actual publish	No sitemap entry until actual publish
Publish	Upsert public search document after commit	Recompute SEO/sitemap eligibility
Update Published	Upsert current public search document after commit	Refresh lastmod/SEO/sitemap eligibility
Archive	Remove public search document	Remove Article URL; archives recalc counts
Trash	Remove public search document	Remove Article URL
Restore to nonpublic state	Remain absent until republished	Remain absent
Permanent Delete	Ensure document absent	Former URL routing follows DOC-09 410; never sitemap
Category/Tag/Author rename/change	Refresh affected indexed taxonomy/author terms	Recompute archive SEO/sitemap as applicable
Site DISABLED	All site docs ineligible or removed	No ordinary public sitemap output
Canonical-domain change	Search text unchanged; destinations reproject current canonical host	Regenerate sitemap/canonical URLs on new host

26.1 Reindex request

18. API15-S-094 accepts a site-scoped reindex command only after search.view + search.reindex.
19. Return 202 AcceptedCommand; do not pretend the rebuild completed synchronously.
20. The mandatory rebuild target is the DOC-16 public search index for exactly one site, reconstructed from current authoritative data. DOC-16 does not require a second mandatory derived Admin index.
21. Admin Post Search may query authoritative persistence directly or use an implementation-specific same-site acceleration/index selected later under DOC-23. If such an Admin acceleration exists, reindex may refresh it as part of the same accepted site-scoped operation, but its existence is not a new DOC-16 product requirement.
22. During rebuild, an existing valid public index may remain queryable only if the implementation can prove tenant-safe consistency; otherwise SearchStatus becomes REINDEXING/DEGRADED according to Section 27.
23. Retries/idempotency/queue mechanics remain DOC-15/18, but tenant identity must remain unchanged across retries.

27. Search Diagnostics and Failure/Degraded Behavior

SearchStatusResponse.state	Meaning
READY	No known site-scoped reindex is active; latest completed index state is available and no known integrity failure requires operator attention.
REINDEXING	A full site-scoped rebuild accepted by API15-S-094 is in progress.
DEGRADED	Search service/index for this site has a known availability or integrity problem, failed rebuild, or source/index drift requiring intervention.

- last_indexed_at is the most recent successful full or authoritative incremental synchronization instant known to the service.
- message is human-readable diagnostics only; it must not disclose another site, raw engine secrets, query DSL or private content.
- Public search transport failure uses DOC-11/15 ERROR/OFFLINE semantics; it must not silently return stale all-site data.

28. Redirect Precedence Integration

DOC-09 remains the route precedence authority. DOC-16 does not reorder it; it defines SEO consequences and reserves crawler resources.

Order	Resolver layer	DOC-16 consequence
1	Host/site resolution	No SEO/search/sitemap/redirect work without one trusted site.
2	Scheme/host/path canonicalization	System 308; no source PageData/SEO body rendered.
2A	DOC-16 crawler resources: /robots.txt, /sitemap.xml, /sitemaps/*	System-generated discovery resources; not Page IDs or manual redirects.
3	Preview namespace	Private/noindex/nofollow; excluded from search/sitemap.
4	Current canonical page route	Live route wins; compute PageData + SEO if eligible.
5	Active manual redirect	301/302 only; source page not rendered/indexed.
6	Historical Post locator	History is evidence only.
7	410 eligibility	NOINDEX system response; no canonical/sitemap.
8	404	NOINDEX system response; no canonical/sitemap.

29. Redirect Create/Edit Conflict and Loop Algorithm

RED16-VAL-01

```
1 authorize redirects.create/edit in CTX-ADMIN-SITE
2 normalize source_path using DOC09 canonical path rules
3 reject source if it is /robots.txt, /sitemap.xml or starts /sitemaps/
4 reject source collision with Home/static/Search/current public entity/Preview namespace (DOC09)
5 validate destination kind/value and status_code in {301,302}
6 INTERNAL_PATH: normalize destination; reject self redirect
7 INTERNAL_PATH: build active same-site redirect graph including proposed edge
8 reject if DFS/topological cycle is found
9 reject if following active internal edges from source would exceed 3 edges
10 if internal destination is site-owned entity, require current public eligibility
11 persist under DOC08 UQ(site_id,source_path); concurrent collision fails
```

29.1 Runtime chain behavior

- Each request reevaluates the current active redirect graph; no redirect grants access to a private target.
- If an INTERNAL_PATH destination becomes unsafe/nonpublic, suppress the redirect and continue DOC-09 404/410 evaluation.

- System 308 canonicalization is not counted as an editorial redirect edge because it is not a redirects row.
- A manual redirect source is excluded from sitemap because the source page is not a canonical content URL.

30. Slug-Change Protection and URL-History Interaction

24. A published Post slug change creates/maintains DOC-07/08 URL history evidence after the durable mutation.
25. URL history alone never redirects; without an active manual redirect the former path remains 404 while the Post exists and 410 only after terminal permanent delete under DOC-09.
26. When the authorized editor has redirects.view + redirects.create, the slug-change workflow may create a protective redirect from old /article/{slug} to the new current /article/{slug}.
27. Protective redirect default status is 301, preserving DOC-09. 302 remains an explicit supported alternative when a manual editor intentionally chooses temporary behavior.
28. The redirect is created only after validating that the old source is no longer a current live route and the new target is currently public-eligible.
29. If the old slug is reused by a different current Post before redirect creation, the new live route wins and the redirect must be rejected.
30. Changing a canonical_override does not create URL history or a redirect because it does not change routing identity.

31. External Redirects and Canonical Interaction

Concern	Rule
Allowed destination	Absolute HTTPS external URL with valid hostname and no embedded credentials, per DOC-09.
Status	301 or 302 only.
Site ownership	Redirect row remains owned by source site; destination does not become site-owned.
Canonical	A redirect source has no rendered canonical tag. Destination page controls its own canonical; DOC-16 does not inject canonical across HTTP redirect.
Search index	Redirect source path is not a public search document.
Sitemap	Redirect source is excluded. External destination is never inserted into the source site sitemap.
Security	Later PDF-19 may block unsafe hosts more strictly but may not make an otherwise private same-site target public.

32. DOC-15 API Integration Map

Operation	DOC-16 usage	No redefinition
GET /api/v1/public/page	Receives path/q/page/page_size and returns DOC-13 PageData with DOC-16-resolved SEO/search order.	Method/path/status/params remain DOC-15.
GET/PATCH /api/v1/admin/sites/{site_key}/seo	Read/edit site SEO template/default/index/follow/social/structured inputs.	Permission seo.view / seo.view+seo.edit; schema remains SiteSeoInput.
Category create/read/update endpoints	Transport category seo_title, meta_description, canonical_override and index_policy through DOC-15 RC5 Category schemas.	SEO fields are omitted without seo.view; supplying any SEO field requires seo.view + seo.edit in addition to the base category create/edit permission. If either SEO permission is absent, reject the whole mutation atomically.
Author create/read/update endpoints	Transport author seo_title, meta_description, canonical_override and index_policy through DOC-15 RC5 Author schemas.	SEO fields are omitted without seo.view; supplying any SEO field requires seo.view + seo.edit in addition to the base author create/edit permission. If either SEO permission is absent, reject the whole mutation atomically.

Operation	DOC-16 usage	No redefinition
GET/POST/PATCH/DELETE redirects endpoints	Transport site-owned redirect CRUD.	301/302/status, idempotency and errors remain DOC-15.
GET .../search/posts	Admin Post search q/status/page/page_size; DOC-16 ranks authorized candidates.	No hidden filters.
GET .../search/internal-links	Same-site internal-link candidate search.	Eligibility remains upstream.
GET .../search/related-posts	Same-site related candidate search.	Automatic-related feature remains CONDITIONAL upstream.
GET .../search/status	Returns READY/REINDEXING/DEGRADED semantics from Section 27.	Schema/status remains DOC-15.
POST .../search/reindex	Accept site-scoped rebuild request.	202/idempotency/rate rules remain DOC-15.

33. Security, Privacy and Non-Disclosure Boundaries

- SEO output never exposes site_id, Admin identities, permissions, revision/autosave data, storage keys, secrets, preview tokens, search debug scores or worker/cache keys.
- Search diagnostics do not disclose other sites or private content names to an unauthorized actor.
- Public search scores are internal and are not returned in SearchPageData.
- Canonical override/redirect validation does not perform SSRF-like content fetching as part of normal save; later security policy may add safe verification through approved infrastructure.
- robots.txt and sitemaps are public discovery resources only for an ACTIVE canonical site and are never treated as authorization controls.
- Unknown hosts never inherit the SEO defaults, sitemap or branding of a default tenant.
- Structured data and breadcrumbs use same-site canonical destinations only except an explicitly validated external canonical URL field; they never pull another tenant object by ID.

34. Edge Cases and Deterministic Examples

Scenario	Required outcome
SITE-01 and SITE-02 both publish /article/future-of-ai	Each resolves, indexes, ranks and sitemaps independently under its own host.
Article explicit seo_title absent	Use Post title, then apply site title template.
Article explicit canonical_override is https://external.example/source	Emit external canonical if valid; keep local route working; exclude local Article URL from sitemap.
Search q=" "	NO_QUERY + articles=[] + no pagination; canonical /search; noindex.
Category exists but has 0 eligible Articles	200 EMPTY page; noindex,follow; exclude from sitemap.
Tag has 3 eligible Articles	Index candidate if site policy allows; route canonical; include base URL in sitemap.
Archive request ?page=2	Serve valid page if results exist; canonical base archive; noindex,follow; never sitemap page 2.
Manual redirect /old -> /article/new; target later archived	Suppress redirect; proceed to 404/410 logic without exposing target state.
Redirect A->B, B->C, C->A proposed	Reject create/edit due cycle.
Redirect chain would become 4 edges	Reject create/edit; maximum remains 3.
Alias host requests sitemap	DOC-09 canonical-host 308 occurs first; canonical host serves sitemap.
Preview fetched successfully	SeoProjection robots false/false, canonical omitted, structured_data=[], no search/sitemap population.

35. Cross-Document Consistency Audit: DOC-01 through DOC-15

DOC-16 was re-audited against the finalized upstream set after the controlled DOC-15 RC5 authorization reconciliation. The Category/Author SEO transport omissions and permission pair remain resolved. DOC-15 RC5 additionally removes the Appendix B schema-registry ambiguity by separating required fields from the complete declared top-level property set and verifies the executable OpenAPI directly. No remaining upstream contradiction or blocking requirement was found for the DOC-16 scope.

Upstream	Result	Finding / consequence
DOC-01	PASS	14-site shared-core/isolated-site model preserved.
DOC-02	PASS after DOC-15 RC5	No new permissions; Category/Author SEO writes now use the frozen seo.view + seo.edit pair plus base entity permission; all SEO/redirect/search authorization references use frozen keys.
DOC-03	PASS	All required SEO/redirect/search capabilities are represented and now have matching DOC-15 transport where required.
DOC-04	PASS	Tenant isolation preserved for search/index/sitemap/canonical/redirect.
DOC-05	PASS	Exactly 14 Page IDs; baseline indexability preserved.
DOC-06	PASS	No new Admin page/module; sitemap remains system effect.
DOC-07	PASS	Uses frozen Post SEO inputs/lifecycle; no new content fields.
DOC-08	PASS	Uses existing Category/Author SEO columns, site_settings.seo_defaults and redirects; no schema changes.
DOC-09	PASS	Route families/system 308/live-route-wins/history/404/410 preserved; DOC-16 crawler resources are non-Page discovery resources.
DOC-10	PASS	SEO/Search/Redirect page actions and handoffs preserved.
DOC-11	PASS	NOINDEX and pagination/query behavior preserved.
DOC-12	PASS	No visual/component redesign.
DOC-13	PASS	Computes exact existing SeoProjection/OpenGraph/Robots/Breadcrumb/Search shapes.
DOC-14	PASS	Same semantics across 14 isolated frontends; no site-to-site import requirement.
DOC-15 RC5	PASS - GAP-01 resolved	CategoryInput/CategoryUpdateInput/CategoryResource expose seo_title, meta_description, canonical_override and index_policy. SEO field read visibility requires seo.view; supplying any SEO field requires seo.view + seo.edit plus the base category create/edit permission, with atomic rejection if either SEO permission is absent.
DOC-15 RC5	PASS - GAP-02 + authorization resolved	AuthorInput/AuthorUpdateInput/AuthorResource expose the same frozen SEO fields with equivalent seo.view + seo.edit conditional write authorization plus base author create/edit permission.
DOC-15 RC5	PASS - schema registry/OpenAPI verification	Appendix B now separately lists required fields and every declared top-level property; Category/Author SEO properties are visible in the registry and match the executable OpenAPI.

AUDIT RESULT

No genuine contradiction requires changing a DOC-16 algorithm. DOC-15 RC5 preserves the resolved Category/Author SEO transport and now explicitly requires seo.view + seo.edit for any SEO-field mutation in addition to the base entity create/edit permission, matching DOC-02/DOC-10. No endpoint, capability, permission key or feature was added; DOC-16 can proceed to approval without this blocker.

36. Traceability and Downstream Dependencies

Trace ID	Requirement chain
TR16-001	DOC-05 Page ID/indexability -> Sections 6,10,11,17 -> QA16-SEO-001..012

Trace ID	Requirement chain
TR16-002	DOC-07 Post SEO inputs -> Sections 5-15 -> QA16-SEO-013..030
TR16-003	DOC-04/08 site isolation -> Sections 4,19-27 -> QA16-ISO-001..010
TR16-004	DOC-09 canonical/redirect route model -> Sections 9,28-31 -> QA16-RED-001..016
TR16-005	DOC-13 SeoProjection/SearchPageData -> Sections 6-17,21-24 -> QA16-DATA-001..012
TR16-006	DOC-15 public/search/SEO/redirect transport -> Sections 22-27,32 -> QA16-API-001..012
TR16-007	DOC-03 FC-SEO-11 -> Sections 18-20 -> QA16-SM-001..014
TR16-008	DOC-03 FC-SEARCH-01..08 -> Sections 21-27 -> QA16-SRCH-001..020

36.1 Downstream authority handoff

Later PDF	May implement/refine	May not redefine
PDF-17 Media/CDN	Public media URLs/variants/delivery used by OG/structured data	Which media fallback wins or cross-site ownership.
PDF-18 Workers/cache	Queues, retry/backoff, incremental indexing/sitemap generation/cache invalidation	DOC-16 eligibility/ranking/canonical/redirect semantics.
PDF-19 Security	Credential/session/security headers, stricter unsafe-host/abuse rules	Relax tenant isolation or make private content indexable.
PDF-20 Scale/ops	Performance targets and operational scaling for search/sitemap	Ranking or eligibility meaning.
PDF-21 QA	Expand executable tests	Weaken acceptance assertions.
PDF-23 Engineering	Select search technology/repository modules	Change logical index fields or dependency direction.
PDF-24 Deployment	Expose crawler resources/proxies/CDN	Change canonical host or path semantics.
PDFs 25-27 Visual	Render SEO warnings/metadata UI and public pages	Change data/algorithm authority.

37. Comprehensive Developer Acceptance Criteria

Acceptance ID	Release-blocking assertion
QA16-SEO-01	Home, Article and five Static valid pages produce SeoProjection with baseline indexability preserved and deterministic resolved title.
QA16-SEO-02	Article title resolves explicit seo_title before Post title and then site template.
QA16-SEO-03	Missing Article meta_description uses excerpt, then approved body summary, then site template; no fabricated marketing copy is created.
QA16-SEO-04	Category title/meta/canonical/index semantics use existing DOC-08 fields when present.
QA16-SEO-05	Tag SEO uses tag name/description plus site defaults and does not assume nonexistent tag SEO columns.
QA16-SEO-06	Author title/meta/canonical/index semantics use existing DOC-08 fields when present.
QA16-SEO-07	Search is always noindex and canonicalizes to clean /search without q/page.
QA16-SEO-08	Preview is noindex,nofollow with canonical omitted and no structured data.
QA16-SEO-09	404 and 410 are noindex and canonical omitted.
QA16-SEO-10	Site NOINDEX cannot be overridden by object INDEX or Page-ID baseline.
QA16-SEO-11	Category/Tag/Author with zero eligible Articles render valid EMPTY but final index=false.
QA16-SEO-12	Archive page>1 is noindex,follow and canonicalizes to the base archive.

Acceptance ID	Release-blocking assertion
QA16-SEO-13	Canonical override is HTTPS, credential-free, fragment-free and cannot point to same-site private/system resources.
QA16-SEO-14	Canonical override never creates a redirect or changes route resolution.
QA16-SEO-15	External canonical override excludes the local route from sitemap eligibility.
QA16-SEO-16	Open Graph title/description/image follow the exact fallback order.
QA16-SEO-17	Missing OG media results in omission, never cross-site media substitution.
QA16-SEO-18	Robots follow resolution obeys site/post policies; Preview always nofollow.
QA16-SEO-19	Configured site title template rejects unknown tokens and must contain {page_title}.
QA16-SEO-20	Explicit valid SEO values are not silently rewritten or keyword-stuffed.
QA16-DATA-01	SeoProjection fields exactly match DOC-13 presence/shape rules.
QA16-DATA-02	OpenGraph.type is ARTICLE for Article and WEBSITE for other ordinary pages.
QA16-DATA-03	BreadcrumbTrail has root-to-current order with exactly one final current item and no destination on current.
QA16-DATA-04	Article breadcrumb_label wins over Article title for current breadcrumb label.
QA16-DATA-05	Generated structured_data is an array and never exposes raw Admin structured_data_inputs.
QA16-DATA-06	Article JSON-LD uses public-safe author/media/canonical/date sources only.
QA16-DATA-07	Search/Preview/404/410 structured_data is [].
QA16-DATA-08	Unknown structured-data source types are rejected instead of passed through.
QA16-DATA-09	Keys beginning with @ cannot be supplied through Admin structured-data inputs.
QA16-DATA-10	Public search results return ArticleSummary projections rather than raw index documents/scores.
QA16-DATA-11	Optional missing SEO fields are omitted rather than implicit null unless DOC-13 says otherwise.
QA16-DATA-12	No site_id, Admin identity, storage key, preview token or search debug score appears in public SEO/search output.
QA16-SM-01	/robots.txt is generated per resolved ACTIVE site and is not a Page ID.
QA16-SM-02	Site NOINDEX produces robots.txt Disallow: / while still declaring canonical sitemap URL.
QA16-SM-03	Normal robots.txt disallows /_preview/ and declares canonical-host sitemap.
QA16-SM-04	/sitemap.xml is a site-local sitemap index and never aggregates all 14 sites.
QA16-SM-05	Only final index=true local-canonical 200 public URLs enter sitemaps.
QA16-SM-06	Search/Preview/404/410 and pagination page>1 never enter sitemaps.
QA16-SM-07	Article with external canonical override is excluded from local sitemap.
QA16-SM-08	Sitemap child resources use fixed type grouping and deterministic URL ascending order.
QA16-SM-09	Each child shard contains at most 25,000 URLs.
QA16-SM-10	Shard numbering starts at 1 and follows sorted order.
QA16-SM-11	lastmod uses authoritative timestamps only and is omitted when not trustworthy.
QA16-SM-12	No changefreq or priority is emitted in v1.
QA16-SM-13	Alias-host crawler-resource requests canonicalize to canonical host first.
QA16-SM-14	Disabled site cannot emit stale ordinary sitemap content.
QA16-SRCH-01	Public search binds site_id before text matching and never returns another site.
QA16-SRCH-02	Blank/whitespace/punctuation-only q returns NO_QUERY + [] with no pagination.
QA16-SRCH-03	Index and query use the same NFKC/case-fold/tokenization rules.

Acceptance ID	Release-blocking assertion
QA16-SRCH-04	No v1 stemming/transliteration/synonym/wildcard/operator semantics are applied silently.
QA16-SRCH-05	Only PUBLISHED/public-eligible Posts enter public search index.
QA16-SRCH-06	Post NOINDEX may still appear in site-internal public search if otherwise public-eligible.
QA16-SRCH-07	Ranking contributions match Section 23 exactly.
QA16-SRCH-08	Tie order is score DESC, first_published_at DESC, post_id ASC.
QA16-SRCH-09	Public result order is preserved into DOC-13 ArticleSummary array.
QA16-SRCH-10	Public response never exposes internal score.
QA16-SRCH-11	DOC-15 page default=1 and page_size default=20/max50 are respected.
QA16-SRCH-12	Changing query resets page=1.
QA16-SRCH-13	Admin Post search requires posts.view in trusted site context.
QA16-SRCH-14	Admin search status filter cannot broaden site scope.
QA16-SRCH-15	Admin ranking gives exact title then exact slug highest priority.
QA16-SRCH-16	Internal-link/related candidate search remains same-site and cannot invent eligibility.
QA16-SRCH-17	Reindex requires search.view + search.reindex and returns 202 under DOC-15.
QA16-SRCH-18	Reindex is site-scoped and rebuildable from authoritative data.
QA16-SRCH-19	SearchStatus READY/REINDEXING/DEGRADED meanings match Section 27.
QA16-SRCH-20	Search failure never falls back to mixed/all-site stale results.
QA16-RED-01	System 308 canonicalization remains before page/manual redirect processing.
QA16-RED-02	Current canonical public route wins over any manual redirect source collision.
QA16-RED-03	/robots.txt, /sitemap.xml and /sitemaps/* cannot be manual redirect sources.
QA16-RED-04	Manual redirect statuses are only 301/302.
QA16-RED-05	Redirect source normalization matches DOC-09 request normalization.
QA16-RED-06	INTERNAL_PATH self-redirect is rejected.
QA16-RED-07	Any active internal redirect cycle is rejected.
QA16-RED-08	Any proposed chain exceeding 3 manual edges is rejected.
QA16-RED-09	Internal target must be currently public-eligible at commit time.
QA16-RED-10	Runtime unsafe/nonpublic internal destination suppresses redirect and proceeds to 404/410.
QA16-RED-11	External destination is HTTPS and credential-free.
QA16-RED-12	Redirect source never appears in XML sitemap.
QA16-RED-13	URL history alone never produces redirect.
QA16-RED-14	Published slug change may create protective redirect only with required redirect permission.
QA16-RED-15	Protective redirect default is 301.
QA16-RED-16	Reused old slug occupied by a new live Post blocks protective redirect creation.
QA16-ISO-01	Identical query/slug/source path on two sites remains fully isolated.
QA16-ISO-02	Canonical generation uses owning site canonical domain only.
QA16-ISO-03	Site defaults never apply to another tenant.
QA16-ISO-04	Structured data never mixes another tenant author/media/navigation.

Acceptance ID	Release-blocking assertion
QA16-ISO-05	Sitemap output contains only canonical-domain URLs for that site.
QA16-ISO-06	Reindex job retains original site identity through acceptance/retry handoff.
QA16-ISO-07	Admin search cannot reveal wrong-site drafts via valid UUID/title.
QA16-ISO-08	Redirect lookup key includes site identity + normalized source path.
QA16-ISO-09	Unknown host never receives another site robots/sitemap/SEO output.
QA16-ISO-10	Cross-site isolation failure is release-blocking.
QA16-API-01	DOC-16 implementation uses API15-PUB-001 without adding public discovery endpoints.
QA16-API-02	SEO edit uses API15-S-081 with seo.view + seo.edit and existing SiteSeoInput.
QA16-API-03	Redirect CRUD uses API15-S-082..086 without changing methods/status codes.
QA16-API-04	Admin search uses API15-S-090 q/status/page/page_size only.
QA16-API-05	Internal-link/related/status endpoints remain API15-S-091..093.
QA16-API-06	Reindex remains API15-S-094 202 AcceptedCommand.
QA16-API-07	DOC-15 idempotency/rate/error semantics remain unchanged.
QA16-API-08	Category SEO transport is present in DOC-15 RC5 for seo_title, meta_description, canonical_override and index_policy, with seo.view read visibility and seo.view + seo.edit mutation gating in addition to the base entity permission.
QA16-API-09	Author SEO transport is present in DOC-15 RC5 for seo_title, meta_description, canonical_override and index_policy, with seo.view read visibility and seo.view + seo.edit mutation gating in addition to the base entity permission.
QA16-API-10	No new API request property is accepted solely because DOC-16 needs an algorithm input.
QA16-API-11	Public SeoProjection response shape remains DOC-13/OpenAPI schema compatible.
QA16-API-12	DOC-15 RC5 Category/Author SEO transport preserves DOC-16 field semantics without changing Page IDs, routes, capabilities or permission keys.

RELEASE GATE

104 DOC-16 acceptance assertions are normative. Any tenant leakage, route/canonical contradiction, sitemap contamination, wrong search order, hidden API expansion, unsafe redirect or Page-ID/indexability violation is release-blocking. The former GAP-01/GAP-02 transport blockers and the RC3 SEO permission mismatch are resolved by DOC-15 RC5 and are now positive contract assertions.

Appendix A - Canonical 14-Site Discovery Registry

Site key	Canonical public domain	DOC-16 rule
SITE-01	WhatsYourRashi.in	Independent SEO/search/sitemap/redirect namespace; same algorithms, isolated data.
SITE-02	ApkiRashi.com	Independent SEO/search/sitemap/redirect namespace; same algorithms, isolated data.
SITE-03	DailyRashi360.com	Independent SEO/search/sitemap/redirect namespace; same algorithms, isolated data.
SITE-04	ZodiacDaily.in	Independent SEO/search/sitemap/redirect namespace; same algorithms, isolated data.
SITE-05	DailyHoroscope360.com	Independent SEO/search/sitemap/redirect namespace; same algorithms, isolated data.
SITE-06	MyHoroscope360.com	Independent SEO/search/sitemap/redirect namespace; same algorithms, isolated data.

Site key	Canonical public domain	DOC-16 rule
SITE-07	AstroToday360.com	Independent SEO/search/sitemap/redirect namespace; same algorithms, isolated data.
SITE-08	ApkiKundli.com	Independent SEO/search/sitemap/redirect namespace; same algorithms, isolated data.
SITE-09	MatchYourKundli.com	Independent SEO/search/sitemap/redirect namespace; same algorithms, isolated data.
SITE-10	ApkaNakshatra.com	Independent SEO/search/sitemap/redirect namespace; same algorithms, isolated data.
SITE-11	JaniyePanchang.com	Independent SEO/search/sitemap/redirect namespace; same algorithms, isolated data.
SITE-12	RashiRemedies.com	Independent SEO/search/sitemap/redirect namespace; same algorithms, isolated data.
SITE-13	ApkaPandit.in	Independent SEO/search/sitemap/redirect namespace; same algorithms, isolated data.
SITE-14	VastuShastra360.in	Independent SEO/search/sitemap/redirect namespace; same algorithms, isolated data.

Appendix B - Public Page ID SEO Matrix

Page ID	Canonical route	Baseline	DOC-16 final rule
PUB-HOME-01	/	INDEX-ELIGIBLE	See Sections 10-12/17
PUB-ARTICLE-01	/article/{post_slug}	INDEX-ELIGIBLE	See Sections 10-12/17
PUB-CATEGORY-01	/category/{category_slug}	CONDITIONAL-INDEX	See Sections 10-12/17
PUB-TAG-01	/tag/{tag_slug}	CONDITIONAL-INDEX	See Sections 10-12/17
PUB-AUTHOR-01	/author/{author_slug}	CONDITIONAL-INDEX	See Sections 10-12/17
PUB-SEARCH-01	/search	NOINDEX	See Sections 10-12/17
PUB-STATIC-ABOUT-01	/about	INDEX-ELIGIBLE	See Sections 10-12/17
PUB-STATIC-CONTACT-01	/contact	INDEX-ELIGIBLE	See Sections 10-12/17
PUB-STATIC-PRIVACY-01	/privacy-policy	INDEX-ELIGIBLE	See Sections 10-12/17
PUB-STATIC-TERMS-01	/terms-of-use	INDEX-ELIGIBLE	See Sections 10-12/17
PUB-STATIC-DISCLAIMER-01	/disclaimer	INDEX-ELIGIBLE	See Sections 10-12/17
PUB-SYS-PREVIEW-01	/_preview/post/{post_id}	NOINDEX	See Sections 10-12/17
PUB-SYS-404-01	No direct route	NOINDEX	See Sections 10-12/17
PUB-SYS-410-01	No direct route	NOINDEX	See Sections 10-12/17

Appendix C - Search Score Reference

Signal	Score contribution
Normalized full query equals title_text	+160
Normalized full query appears as contiguous phrase in title_text	+110
All unique query tokens appear in title_text	+70
title token coverage ratio	+40 x coverage

Signal	Score contribution
Full query phrase in seo_title_text	+45
seo_title token coverage	+20 x coverage
Full query phrase in excerpt_text	+30
excerpt token coverage	+18 x coverage
category/tag token coverage	+18 x max(category_coverage, tag_coverage)
author token coverage	+14 x coverage
slug token coverage	+10 x coverage
body token coverage	+8 x coverage

Tie-breaker: score DESC -> first_published_at DESC NULLS LAST -> post_id ASC. No cross-site signal, personalization or paid boost exists in v1.

Appendix D - Sitemap Resource Registry

Resource	Purpose	Contents
/sitemap.xml	Sitemap index	References non-empty site-local child sitemaps only.
/sitemaps/pages.xml	Fixed/indexable non-entity pages	Home + eligible fixed static URLs.
/sitemaps/articles-{n}.xml	Article shards	Final sitemap-eligible Article canonical route URLs.
/sitemaps/categories-{n}.xml	Category archive shards	Final indexable base Category URLs.
/sitemaps/tags-{n}.xml	Tag archive shards	Final indexable base Tag URLs.
/sitemaps/authors-{n}.xml	Author archive shards	Final indexable base Author URLs.

Appendix E - Redirect Decision Matrix

Situation	Outcome
HTTP/alias/noncanonical safe path	DOC-09 system 308 before page body.
Current public route exists	Render current route; redirect row cannot override.
Crawler resource path	Serve DOC-16 system resource; manual redirect source prohibited.
No live route; active safe manual redirect	301/302 Location.
Internal redirect target unsafe at runtime	Suppress redirect; continue history/404/410.
No redirect; former public slug; Post exists/nonpublic	404.
No redirect; former public slug; Post permanently deleted	410.
No route/history	404.

Appendix F - One-Page Developer Execution Reference

REQUEST ORDER

1) DOC-09 canonicalize host/path. 2) Bind trusted SiteContext. 3) Resolve current page / crawler resource / manual redirect according to authority. 4) Enforce public eligibility. 5) Compute DOC-16 SEO/search/sitemap decision. 6) Project exact DOC-13 data. 7) Transport through DOC-15. Never invert this order.

Concern	Do
SEO title	Explicit object title -> visible title -> site {page_title} template.
Meta	Explicit -> approved page summary -> safe Article body summary -> site template -> omit.
Canonical	Valid supported override else DOC-09 route; Search clean /search; Preview/404/410 omit.
Robots	Hard Page-ID gates first; then site/object policy; conditional archives need >=1 eligible Article; page>1 noindex.
OG	Article explicit OG -> resolved SEO/meta; image OG -> featured -> site default.
Structured data	Generate allowlisted WebSite/Article/CollectionPage/ProfilePage/Person/WebPage/BreadcrumbList only.
Sitemap	Only index=true + local route-canonical + 200 eligible URLs; 25k/shard; no pagination/search/preview/system.
Public search	NFKC/casefold/tokenize; same-site PUBLISHED only; exact ranking formula; 20 default/50 max.
Reindex	Site-scoped 202 command; rebuild derived index; no synchronous completion claim.
Redirect	Preserve DOC-09 live-route-wins and max-3 chain; crawler resources reserved; no unsafe target.
Upstream reconciliation	DOC-15 RC5 resolves Category/Author SEO transport and authorization: read requires seo.view; any SEO-field write requires seo.view + seo.edit plus the base entity create/edit permission; atomic failure on missing SEO authority.